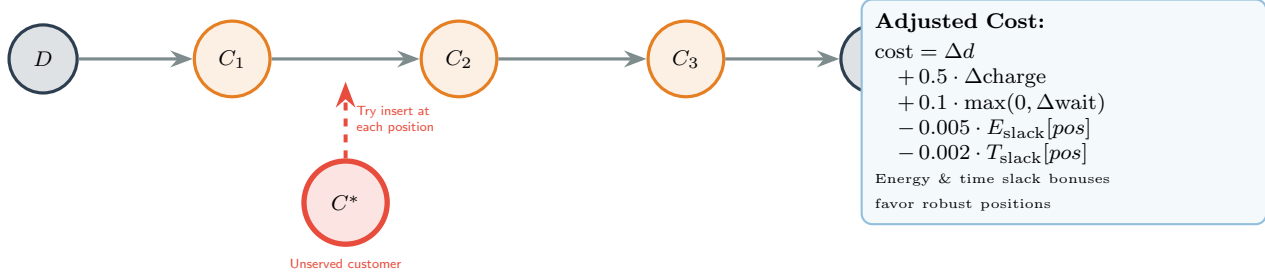


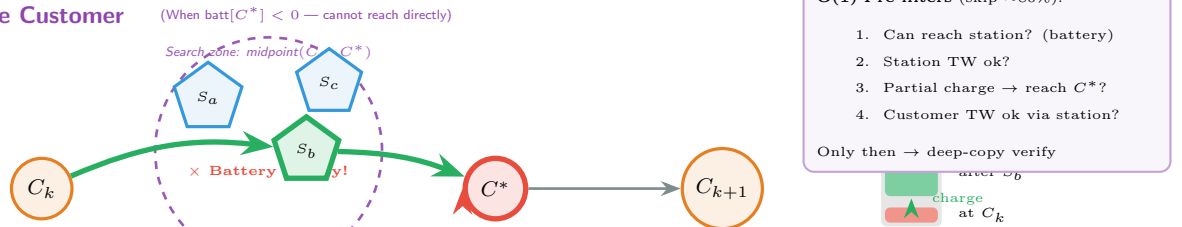
SmartTimeAwareStationRepair Operator

Two-Phase Insertion with Free Charge Detection & Multi-Zone Station Search

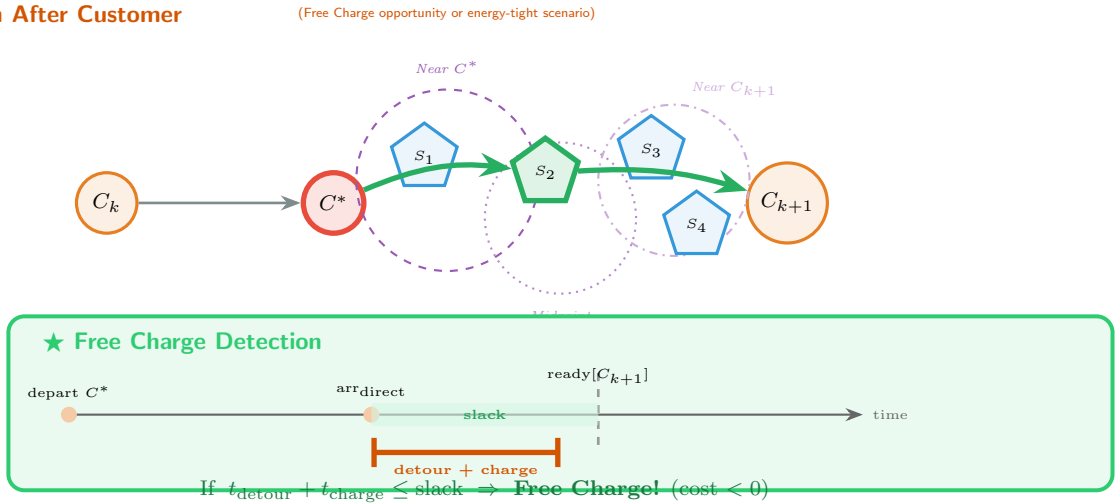
Phase 1: Direct Insertion



Phase 2a: Station Before Customer



Phase 2b: Station After Customer



Cost Decision Logic

Free Charge:

$$\text{cost} = -E_{\text{needed}}$$

Always negative $\Rightarrow$

highest priority. Energy gained for “free” (hidden in wait time).

Normal Station:

$$\text{cost} = \Delta d + t_{\text{charge}}$$

$$- 0.5 \cdot E_{\text{needed}}$$

Detour + charge cost minus energy value (reduces risk of future station insertion).

Direct (no station):

Legend

- D Depot
- C Customer
- C^* New insert
- S Station
- S Chosen
- Search zone